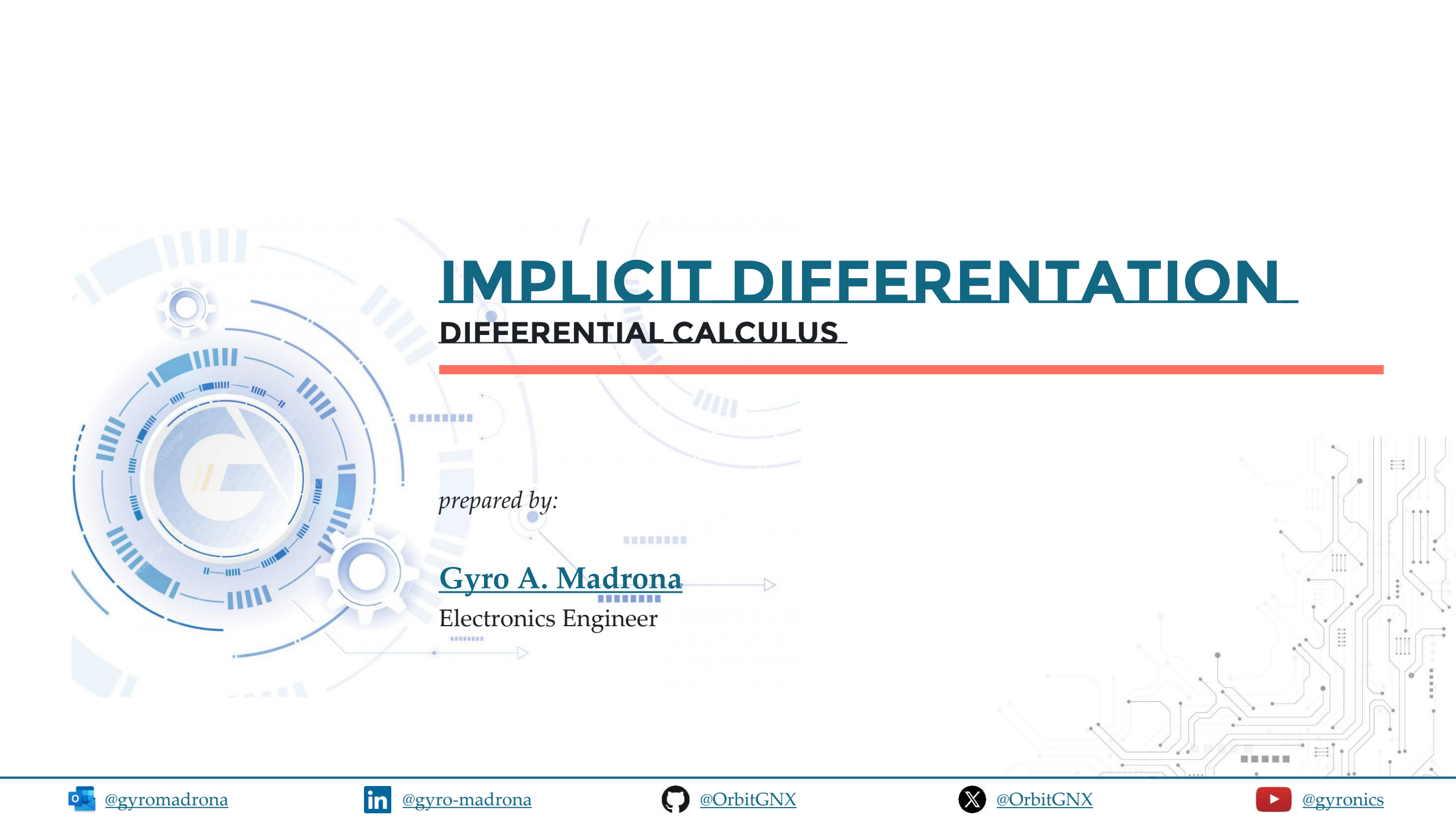




IMPLICIT DIFFERENTIATION

DIFFERENTIAL CALCULUS

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TOPIC OUTLINE

Implicit Differentiation



IMPLICIT DIFFERENTIATION



EXPLICIT FUNCTION

example

$$y = 2x + 3$$

$$y = \sin x$$

The dependent variable (y) is expressed directly in terms of the independent variable (x).

Form

$$y = f(x)$$



IMPLICIT FUNCTION

The relationship between the dependent variable (y) and independent variable (x) is not directly solved for one variable in terms of the other. Instead, the relationship is expressed as an equation involving both variables.

Form

$$F(x, y) = 0$$

example

$$2x - y + 3 = 0$$

$$x^3 + y^3 = 6xy$$



EXERCISE

If $x^2 + y^2 = 25$, find $\frac{dy}{dx}$.

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(25) \rightarrow 0$$

$$2x + 2y \frac{dy}{dx} = 0$$

$$y \frac{dy}{dx} = -x$$

$$\boxed{\frac{dy}{dx} = -\frac{x}{y}}$$

ans

Solution

$$f = y^2, \text{ let } u = y$$

$$f' = \frac{df}{du} \cdot \frac{du}{dx}$$

$$f' = \frac{d}{du} u^2 \cdot \frac{du}{dx}$$

$$f' = 2u \cdot \frac{du}{dx}$$

$$f' = 2y \frac{dy}{dx}$$



EXERCISE

Find y' if $x^3 + y^3 = 6xy$

$$\frac{d}{dx}(x^3) + \frac{d}{dx}(y^3) = \frac{d}{dx}(u \cdot v)$$

$$3x^2 + 3y^2 \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}(ux)$$

$$3x^2 + 3y^2 \frac{dy}{dx} = 6x \frac{dy}{dx} + 6y$$

$$3y^2 \frac{dy}{dx} - 6x \frac{dy}{dx} = 6y - 3x^2$$

$\div 3$

Solution

$$f = y^3, \text{ let } u = y$$

$$f' = \frac{df}{du} \cdot \frac{du}{dx}$$

$$f' = \frac{d}{du} u^3 \cdot \frac{du}{dx}$$

$$f' = 3u^2 \cdot \frac{du}{dx}$$

$$f' = 3y^2 \frac{dy}{dx}$$



EXERCISE

Find y' if $x^3 + y^3 = 6xy$

Solution

$$\frac{dy}{dx} \frac{(y^2 - 2x)}{y^2 - 2x} = \frac{2y - x^2}{y^2 - 2x}$$

$$\frac{dy}{dx} = \frac{2y - x^2}{y^2 - 2x}$$

ans



EXERCISE

Find y' if $\cos x + \sqrt{y} = 5$.

Solution

$$\frac{d}{dx}(\cos x) + \frac{d}{dx}(\sqrt{y}) = \frac{d}{dx}(5)$$

$$-\sin x + \frac{1}{2\sqrt{y}} \cdot \frac{dy}{dx} = 0$$

$$\frac{1}{2\sqrt{y}} \cdot \frac{dy}{dx} = \sin x$$

$$\frac{dy}{dx} = 2\sqrt{y} \sin x$$

ans



EXERCISE

If $\cos(xy) = 1 + \sin y$, find $\frac{dy}{dx}$.

$$\frac{d}{dx}(\cos^{n}xy) = \cancel{\frac{d}{dx}(1)} + \frac{d}{dx}(\sin y)$$

$$-(x \frac{dy}{dx} + y) \sin xy = \cos y \cdot \frac{dy}{dx}$$

$$-x \sin(xy) \cdot \frac{dy}{dx} - y \sin(xy) = \cos(y) \cdot \frac{dy}{dx}$$

$$\cos(y) \frac{dy}{dx} + x \sin(xy) \frac{dy}{dx} = y \sin(xy)$$

Solution

$$f = \cos xy, \text{ let } u = xy$$

$$f' = \frac{df}{du} \cdot \frac{du}{dx}$$

$$f' = \frac{d}{du} \cos u \cdot \frac{d}{dx}(xy)$$

$$f' = -\sin u \cdot x \frac{dy}{dx} + y \cancel{\frac{dx}{dx}}^1$$

$$f' = -(x \frac{dy}{dx} + y) \sin xy$$



EXERCISE

If $\cos(xy) = 1 + \sin y$, find $\frac{dy}{dx}$.

Solution

$$\frac{dy}{dx} [\cos(y) + x \sin(xy)] = y \sin(xy)$$

$$\frac{dy}{dx} = \frac{y \sin(xy)}{\cos(y) + x \sin(xy)}$$

ans



EXERCISE

Use implicit differentiation to find the equation of the

Solution

tangent line to the curve $x^2 + 2xy - y^2 + x = 2$

@(1,2).

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(2xy) - \frac{d}{dx}(y^2) + \frac{d}{dx}(x) = \frac{d}{dx}(2)$$

$$2x + 2x \frac{dy}{dx} + y \frac{d}{dx} 2x - 2y \frac{dy}{dx} + 1 = 0$$

$$2x + 2x \frac{dy}{dx} + 2y - 2y \frac{dy}{dx} + 1 = 0$$

$$\frac{dy}{dx}(2x - 2y) = -2x - 2y - 1 \leftarrow -(2x + 2y + 1)$$



EXERCISE

Use implicit differentiation to find the equation of the **tangent line** to the curve $x^2 + 2xy - y^2 + x = 2$ @ (1,2).

$$\frac{dy}{dx} = - \frac{2x + 2y + 1}{2x - 2y} \leftarrow \text{slope}$$

@ (1,2)

$$\frac{dy}{dx} = - \frac{2(1) + 2(2) + 1}{2(1) - 2(2)}$$

$$\frac{dy}{dx} = \frac{7}{2} \leftarrow \text{slope of tangent line @ (1,2)}$$

Solution

point-slope form

$$y - y_1 = m(x - x_1)$$

$$y - 2 = \frac{7}{2}(x - 1)$$

$$y = \frac{7}{2}x - \frac{7}{2} + 2$$

$$y = \frac{7}{2}x - \frac{3}{2}$$

ans



EXERCISE

Find y' for $y \sin(x^2) = x \sin(y^2)$ by implicit differentiation.

Solution

$$y \frac{d}{dx}(\sin x^2) + \sin x^2 \frac{d}{dx}(y) = x \frac{d}{dx}(\sin y^2) + \sin y^2 \frac{d}{dx}(x)$$

let $u = y^2$

$$y(2x \cos x^2) + \sin x^2 \cdot \frac{dy}{dx} = x(\cos y^2 \cdot 2y \frac{dy}{dx}) + \sin y^2$$

$$2xy \cos(x^2) + \sin(x^2) \frac{dy}{dx} = 2xy \cos(y^2) \frac{dy}{dx} + \sin(y^2)$$

$$\frac{dy}{dx} [\sin(x^2) - 2xy \cos(y^2)] = \sin(y^2) - 2xy \cos(x^2)$$



EXERCISE

Find y' for $y \sin(x^2) = x \sin(y^2)$ by implicit differentiation.

Solution

$$\frac{dy}{dx} = \frac{\sin(y^2) - 2xy \cos(y^2)}{\sin(x^2) - 2xy \cos(x^2)}$$

ans



EXERCISE

Find y' for $\tan(x/y) = x + y$ by implicit differentiation.

Solution

$$\frac{d}{du}(\tan u) \cdot \frac{d}{dx}\left(\frac{x}{y}\right) = \frac{d}{dx}(x) + \frac{d}{dx}(y)$$

$$\sec^2 u \cdot \frac{y \frac{dx}{dx} - x \frac{dy}{dx}}{y^2} = 1 + \frac{dy}{dx}$$

$$\sec^2 u \left(y - x \frac{dy}{dx} \right) = y^2 \left(1 + \frac{dy}{dx} \right)$$

$$y \sec^2 u - x \sec^2 u \frac{dy}{dx} = y^2 + y^2 \frac{dy}{dx}$$



EXERCISE

Find y' for $\tan(\frac{x}{y}) = x + y$ by implicit differentiation.

Solution

$$\frac{dy}{dx} (y^2 + x \sec^2 \frac{x}{y}) = y \sec^2 \frac{x}{y} - y^2$$

$$\frac{dy}{dx} = \frac{y \sec^2(\frac{x}{y}) - y^2}{x \sec^2(\frac{x}{y}) + y^2}$$

ans



EXERCISE

Use implicit differentiation to find the equation of the

Solution

tangent line to the curve $y \sin(2x) = x \cos(2y)$

@ $(\pi/2, \pi/4)$.

$$y \frac{d}{dx}(\sin 2x) + \sin 2x \frac{d}{dx}(y) = x \frac{d}{dx}(\cos 2y) + \cos 2y \frac{d}{dx}(x)$$

$$y(2 \cos 2x) + \sin 2x \frac{dy}{dx} = x(-2 \sin 2y \cdot \frac{dy}{dx}) + \cos 2y$$

$$2y \cos(2x) + \sin(2x) \frac{dy}{dx} = -2x \sin(2y) \frac{dy}{dx} + \cos(2y)$$

$$\frac{dy}{dx} [\sin(2x) + 2x \sin(2y)] = \cos(2y) - 2y \cos(2x)$$



EXERCISE

Use implicit differentiation to find the equation of the **tangent line** to the curve $y \sin(2x) = x \cos(2y)$

@ $(\pi/2, \pi/4)$.

$$\frac{dy}{dx} = \frac{\cos(2y) - 2y \cos(2x)}{\sin(2x) + 2x \sin(2y)}$$

$@(\overset{x}{\pi/2}, \overset{y}{\pi/4})$

$$\frac{dy}{dx} = \frac{\cos(2 \cdot \frac{\pi}{4}) - 2(\frac{\pi}{4}) \cos(2 \cdot \frac{\pi}{2})}{\sin(2 \cdot \frac{\pi}{2}) - 2(\frac{\pi}{2}) \sin(2 \cdot \frac{\pi}{4})}$$

Solution

$$\frac{dy}{dx} = \frac{\frac{\pi}{2}}{\pi}$$

$$\frac{dy}{dx} = \frac{1}{2} \rightarrow \text{slope of tangent line}$$

@ $(\frac{\pi}{2}, \frac{\pi}{4})$



EXERCISE

Use implicit differentiation to find the equation of the **tangent line** to the curve $y \sin(2x) = x \cos(2y)$ @ $(\pi/2, \pi/4)$.

Solution

point-slope form

$$y - y_1 = m(x - x_1)$$

$$y - \frac{\pi}{4} = \frac{1}{2} \left(x - \frac{\pi}{2} \right)$$

$$y = \frac{1}{2}x - \frac{\pi}{4} + \frac{\pi}{4} \rightarrow 0$$

$$y = \frac{1}{2}x$$

ans



LABORATORY

